CHAPTER 1

Sequences and proof by induction

S2, S4, S5, S7, S10, S11, S13

Before you begin · Sequences and index notation

Without a calculator, in six minutes. Answer briefly. Use the "reference sequences" sheet and E1-E2. Then choose the targeted review shown under the questions.

1. Calculate u_1 then u_2 if $u_0 = 3$ and $u_{n+1} = 2u_n - 1$.
2. Distinguish u_{n+1} and $u_n + 1$.
3. Give $1 + 2 + \cdots + 10$.
4. Write $\sum_{k=0}^{3} 2^k$.
5. Determine the multiplier for a decrease of 20%.
6. Give the first three values of `n` in `for n in range(3)`.

Targeted review after the diagnostic check

Redo only the questions you got wrong. For each question: **1.** Substitute the previous value step by step; R01. **2.** Write an index as a subscript and an addition on the line; R01. **3.** Pair the first and last terms; R09 to compare two sums. **4.** Expand the four terms; R08 checks how many there are. **5.** Write 100 minus 20 over 100; R03. **6.** List the values of range before the calculation; R07. The R questions serve as review prompts with the teacher, even if their whole-class routine is planned later.

Objectives

Calculate terms and a sum; write an induction proof; establish monotonicity and bounds; use an auxiliary sequence.

Course notes

Define a sequence and calculate

A real sequence is a function defined on $\mathbb{N}$, or on the integers from some index on n_0. We write u_n for its term of index n. An explicit formula gives u_n directly; a recurrence relation requires also an initial term. For example, $u_0 = 2$ and $u_{n+1} = 3u_n - 1$ give $u_1 = 5$, $u_2 = 14$.

Reference sequences

If $u_{n+1} = u_n + r$, then $u_n = u_p + (n-p)r$. If $u_{n+1} = qu_n$, then $u_n = u_p q^{n-p}$ for $n \geq p$.

$$\sum_{k=p}^{n} u_k = (n-p+1)\frac{u_p + u_n}{2} \quad \text{(arithmetic)}, \qquad \sum_{k=p}^{n} u_k = u_p \frac{1-q^{n-p+1}}{1-q} \quad (q \neq 1).$$

When $q = 1$, the second sum is $(n-p+1)u_p$.

Proof by induction

To prove $P(n)$ for every integer $n \geq n_0$: check $P(n_0)$; fix an arbitrary integer $n \geq n_0$ and assume $P(n)$ true; deduce $P(n+1)$; conclude by the principle of induction. The hypothesis is not the conclusion: it can only be used at the fixed index.

Fully written-up example. Let $u_0 = 1$, $u_{n+1} = 2u_n + 3$. For $n \in \mathbb{N}$, let $P(n)$: "$u_n = 4 \times 2^n - 3$" .

Base case. $4 \times 2^0 - 3 = 1 = u_0$; $P(0)$ is true.

Inductive step. Fix $n \in \mathbb{N}$ and suppose $P(n)$ true. By the definition of the sequence, then by this hypothesis,

$$\begin{aligned} u_{n+1} &= 2u_n + 3 \\ &= 2(4 \times 2^n - 3) + 3 \\ &= 8 \times 2^n - 3 = 4 \times 2^{n+1} - 3. \end{aligned}$$

This is $P(n+1)$: the property is hereditary.

Conclusion. By the principle of induction, $u_n = 4 \times 2^n - 3$ for every $n \in \mathbb{N}$.

Proof: sum of integers. Let $P(n)$: "$\sum_{k=0}^{n} k = \frac{n(n+1)}{2}$ ". For $n = 0$, both sides equal 0: $P(0)$ is true. Fix $n \geq 0$ and suppose $P(n)$ true. Then

$$\begin{aligned} \sum_{k=0}^{n+1} k &= \sum_{k=0}^{n} k + (n+1) \\ &= \frac{n(n+1)}{2} + (n+1) \\ &= \frac{(n+1)(n+2)}{2}. \end{aligned}$$

We proved $P(n+1)$. By induction, the formula holds for every $n \in \mathbb{N}$.

Overall behaviour

The sequence is increasing if $u_{n+1} \geq u_n$ for every index, decreasing if the inequality is reversed. It is bounded above if there is a real number M such that $u_n \leq M$ for all n; bounded below: some m such that $u_n \geq m$; bounded if it is both bounded below and bounded above. To study monotonicity, we can study the sign of $u_{n+1} - u_n$. If all the terms are strictly positive, we can compare $\frac{u_{n+1}}{u_n}$ to 1. The study of a function f such that $u_n = f(n)$ is another possibility.

Arithmetico-geometric sequence

For $u_{n+1} = au_n + b$, $a \neq 1$, solve $\ell = a\ell + b$, then set $v_n = u_n - \ell$. We get $v_{n+1} = av_n$, so

$$u_n = \ell + (u_0 - \ell)a^n, \qquad \ell = \frac{b}{1-a}.$$

If $a = 1$, the sequence is arithmetic: do not divide by $1 - a$.

Example. If $u_0 = 10$ and $u_{n+1} = 0{,}8u_n + 6$, then $\ell = 30$ and $u_n = 30 - 20(0{,}8)^n$. We have $10 \leq u_n < 30$ and $u_{n+1} - u_n = 4(0{,}8)^n > 0$.

Pitfall. Checking a property on a hundred terms does not prove it. An increasing sequence is not necessarily positive; a bounded sequence is not necessarily monotonic.

Understand, try, revisit

1.1 Index, value and update

First use: S2 and S4.

Definition · A sequence is not a finite list

The index n denotes an integer; u_n is a real number. The representative point is $(n; u_n)$. A line connecting the points can help with reading but does not define terms at non-integer indices.

Worked example · Two expressions that look alike

For $u_0 = 2$ and $u_{n+1} = 3u_n - 1$, we get $u_1 = 5$, $u_2 = 14$, $u_3 = 41$. Thus $u_2 + 1 = 15$, but $u_{2+1} = 41$. In a loop, the variable `u` first contains u_n; the instruction `u = 3*u-1` replaces it with u_{n+1}.

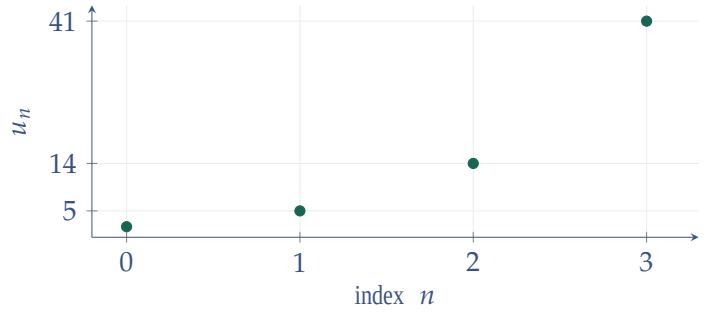


Figure 1.1. Read the term of index 2: its y-coordinate is 14, not 2.

A1 · Immediate application For $v_0 = 1$, $v_{n+1} = 2v_n + 1$, build a table of indices 0 to 3, then place the four points. Calculate separately $v_2 + 1$ and v_{2+1}.

A2 · Partial guidance A loop must calculate $u_0 + u_1 + u_2 + u_3$ for the sequence in the example. It starts at `u=2; total=2`. How many updates are needed? Write the order of the instructions and explain why adding before the update would count twice u_0.

1.2 A local hypothesis, a universal conclusion

First use: S5 and S7.

Worked example · Induction for an inequality

Let $w_0 = 1$ and $w_{n+1} = \frac{1}{2}w_n + 2$. Let us prove $P(n) : 1 \leq w_n < 4$. Initially $w_0 = 1 \in [1; 4[$. Fix $n \in \mathbb{N}$ and suppose $P(n)$. Since $\frac{1}{2} > 0$, $\frac{1}{2} \leq \frac{1}{2}w_n < 2$, so $\frac{5}{2} \leq w_{n+1} < 4$. Since $\frac{5}{2} \geq 1$, this implies $P(n+1)$. The principle of induction gives the bounds at every index.

A3 · Proof to be completed For $a_0 = 2$, $a_{n+1} = 2a_n + 1$, complete: " $P(n)$ means $a_n = 3 \times 2^n - 1$. At index 0, . . . Fix . . . and assume . . . Then $a_{n+1} = \ldots = 3 \times 2^{n+1} - 1$. Hence . . ."

A4 · Fix a proof A student writes: "If $u_n \geq 2$, then $u_{n+1} = 2u_n \geq 4 \geq 2$. So all the terms are at least 2." He did not give u_0. Construct a counterexample, then state which data would complete the proof.

1.3 Count before summing

First use: S10.

Property · Number of terms

For integers $p \leq n$, the sum $\sum_{k=p}^{n} u_k$ contains $n - p + 1$ terms. This can be checked by writing out the first few and the last, before any formula.

Worked example · A shifted geometric sum

In $\sum_{k=3}^{7} 2^k$, there are five terms and the first equals 8. The sum is $8 \frac{1-2^5}{1-2} = 248$. For prove the formula without memorising it in isolation, write $S = 1 + q + \cdots + q^n$, then subtract $qS = q + \cdots + q^{n+1}$: $(1-q)S = 1 - q^{n+1}$. Dividing assumes $q \neq 1$.

A5 · Independent proof Prove by induction, for every $n \in \mathbb{N}$, $\sum_{k=0}^{n}(2k+1) = (n+1)^2$. Explain why the term added in the inductive step is $2 n + 3$, and not $2n + 1$.

1.4 Recognise a structure without confusing it with a limit

First use: S11 and S13.

Worked example · Choosing the shift

For $w_{n+1} = \frac{1}{2}w_n + 2$, the fixed value is 4. Subtracting 4 from both sides, $w_{n+1} - 4 = \frac{1}{2}(w_n - 4)$. Thus $w_n = 4 - 3\,2^{-n}$. The shift is not arbitrary: it removes the constant term. This explicit calculation does not yet assume a convergence theorem.

A6 · Choice of strategy A subscription costs 40 euros in the first year, then its cost increases each year by 5 euros. Another costs 40 euros and then increases by 10 % per year. Write both sequences, compare the first three years and calculate the cumulative costs over ten years. Justify the number of terms.

A first step: First name the type of the two changes. The tenth year corresponds to index 9 if the first is index 0.

One minute, spoken aloud. Explain the difference between "for every n, $u_n \leq 4$" and "there exists n such that $u_n \leq 4$". Give a sequence satisfying the second but not the first.

Delayed review · S20 then S41. Revisit $w_n = 4 - 3\,2^{-n}$ to prove its convergence; in probability, look again for the shift when a probability satisfies an affine relation.

Investigation

A tank topped up every day - 35 min.

A tank initially contains 100 litres of water. Each day, 20 % of the volume present is removed, then 30 litres are added. For every natural number n, let u_n the volume of water, in litres, after these two operations performed n times. Thus, $u_0 = 100$.

1. Calculate u_1 and u_2, then express u_{n+1} as a function of u_n.
2. Which value of the volume would remain unchanged after one day? Prove by induction that, for all $n \in \mathbb{N}$, $100 \leq u_n < 150$.

3. For every $n \in \mathbb{N}$, set $v_n = u_n - 150$. Show that (v_n) is geometric. Deduce the expressions for v_n and of u_n as a function of n.

4. Write a Python function with no argument that returns the first integer n for which $u_n > 145$. Run the program and interpret the result.

5. A student calculated the first hundred terms and notices that they are less than 150. Is this observation enough to prove that all the terms are? Justify.

Progressive exercises

E1 · Direct application. Consider the sequence (u_n) defined, for every natural number n, by $u_n = 5 - 3n$. Calculate u_4, then the sum $\sum_{k=0}^{20} u_k$.

E2 · Application and justification. Let (v_n) be defined by $v_0 = 3$ and, for every natural number n, $v_{n+1} = 2v_n$. Express v_n as a function of n, then calculate $\sum_{k=0}^{n} v_k$.

E3 · Method to organise. Prove by induction that, for every natural number n, $3^n \geq 2n + 1$.

E4 · Choice of strategy. Consider the sequence (u_n) defined on $\mathbb{N}$ by $u_n = \frac{n+1}{n+2}$. Prove that it is increasing and give bounds valid for all its terms.

E5 · Synthesis or proof. Prove by induction that, for every natural number n, $8^n - 1$ is divisible by 7. Congruences will not be used.

Work from the textbook or a past paper

Barbazo, p. 28, Exercises 22 and 25. In class, in S5: exercise 22, to express properties of sequences with a quantifier. At home after S11: exercise 25 in full, to link induction, upper bound and monotonicity. Write the base case, inductive step and conclusion separately.

Homework

DM1 - A sum and a threshold - 45 min, after S11.

Consider the sequence (u_n) defined by $u_0 = 2$ and, for every $n \in \mathbb{N}$, $u_{n+1} = \frac{1}{2}u_n + 3$.

1. Prove by induction that $2 \leq u_n < 6$ for all $n \in \mathbb{N}$. Deduce the direction of variation of (u_n).

2. Let $v_n = u_n - 6$ for all $n \in \mathbb{N}$. Show that (v_n) is geometric, then express u_n in function of n.

3. For every $n \in \mathbb{N}$, let $S_n = \sum_{k=0}^{n} u_k$. Express S_n as a function of n.

4. Using the explicit formula and powers of 2, determine the smallest integer n such that $u_n > 5.99$. Check that the previous integer does not work.

5. Express $6 - \frac{S_n}{n+1}$ and justify that this difference is strictly positive. Interpret the result in considering the mean of the first $n + 1$ terms; no limit calculation is required.

MCQ A · consolidation (after S7)

After S7: first induction proofs.

For each question, only one option is correct. Justify your choice orally before looking at the solutions.

1. If $u_0 = 2$ and $u_{n+1} = u_n + 3$, then u_2 is:

A. 5 B. 8

C. 11 D. 2^4

2. To prove $P(n)$ by induction from 0, the inductive step consists of:

A. assume $P(n+1)$ B. only check $P(0)$

C. assume $P(n)$ then prove $P(n+1)$ D. solve $P(n) = 0$

3. The sum $\sum_{k=0}^{4} 2^k$ includes:

A. 4 terms B. 5 terms

C. 8 terms D. 2^4 terms

4. If $u_{n+1} - u_n > 0$ for every n, then:

A. u is bounded above B. u is increasing

C. u converges D. $u_n > 0$

5. In `while u <= 10: u = u + 1`, the loop stops when:

A. $u = 10$ B. $u < 10$

C. $u > 10$ D. $u < 0$

MCQ B · review and choice of method (after S20)

After S20: monotone convergence and auxiliary sequence.

For each question, only one option is correct. Justify your choice orally before looking at the solutions.

6. A sequence is increasing and bounded above by 7. We can conclude that it:

A. tends to 7 B. converges

C. is positive D. is geometric

7. For $u_{n+1} = 0{,}5u_n + 4$, any equilibrium value satisfies:

A. $L = 0{,}5L + 4$ B. $L = u_0$

C. $L = 0{,}5 + 4L$ D. $L = 4^n$

8. If $v_n = u_n - 8$ and $u_{n+1} = 0{,}5u_n + 4$, then:

A. $v_{n+1} = v_n + 4$ B. $v_{n+1} = 0{,}5v_n$

C. $v_{n+1} = 8v_n$ D. $v_{n+1} = u_n$

9. "u_n belongs to $]5{,}99; 6{,}01[$" is equivalent to:

A. $u_n < 6$ B. $|u_n - 6| < 0{,}01$

C. $|u_n| < 6$ D. $u_n > 0{,}01$

10. A simulation of 20 terms proves that:

A. the statement is true for every n B. the sequence converges

C. the 20 calculated terms have the observed property D. induction is unnecessary

True or false · justification expected. For each statement, indicate whether it is true or false, then write a proof or a counterexample.

1. Every increasing sequence converges.

2. If $u_n = 6 - 2^{-n}$, then $u_n < 6$ for all n.

3. Solve $L = f(L)$ is enough to prove that every sequence $u_{n+1} = f(u_n)$ converges.

Review of prior work

I can compute a term, a sum, set up an induction proof and study monotonicity using $u_{n+1} - u_n$.
To watch out for: confusing an index with a value; turning a numerical observation into a proof.